

VO NGOC KHA AI

CHEMICAL ENGINEERING GRADUATE — QA/QC & R&D

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Final-year Chemical Technology student at IUH — interned at a paint plant, hands-on with coating film test instruments in a QA/QC workflow.



PROFILE

Final-year **Chemical Engineering Technology** student at Industrial University of Ho Chi Minh City (IUH), with a focus on **organic chemistry**. Completed an internship at **Son Hai Van Paint Co., Ltd.**, covering the full paint production chain — raw materials (alkyd, acrylic, epoxy and polyester resins; solvents; pigments and extenders), paste dispersion, fine grinding, let-down, and final quality control. Operated coating film test instruments (viscosity, hiding power, fineness, density, gloss, adhesion) and read results against standardised inspection forms. Thesis work focuses on **leather-like composite materials from water hyacinth and agricultural by-products**. Able to use standard measurement and basic analytical instruments in a university laboratory. Careful and accountable with test data, and used to the rhythm of shift work in a plant. Seeking a **QA/QC or R&D** position in paint, coatings, materials or industrial chemicals.

EXPERIENCE

Intern — Technical Department (QA/QC)

📅 Apr – Jun 2026

Son Hai Van Paint Co., Ltd. — Supervisor: Le Quoc Thoi, Technical Manager

📍 Tay Ninh

- **Standards-based quality control:** carried out sampling and inspection across the **IQC — IPQC — FQC** stages on **400–500 kg** industrial batches; recorded results on standardised forms under an **ISO 9001:2015** system.
- **Operated coating film test instruments:** Stormer viscometer (KU), Pfund Cryptometer hiding power, Hegman fineness gauge, density cup, gloss meter (20°, 60°, 85°) and adhesion tester — measured at **25 ± 0.5 °C** since KU viscosity is temperature-sensitive, and judged pass/fail against technical tolerances (e.g. density **1.25 ± 0.02 g/cm³**, alkyd primer ~70 KU).
- **Familiar with long-duration durability testing** in the plant laboratory: **QUV** accelerated weathering, cylindrical mandrel bend (TCVN 2099/ISO 1519), impact resistance (TCVN 2100-2/ISO 6272-2) and **5% NaCl salt spray at 35 °C for 1000 h** (ISO 7253) — including specimen preparation and rating of rust and blistering (ISO 4628-2) and scribe creep (ASTM D1654).
- **Colour and appearance:** used a colour matching cabinet and spectrophotometer to compare batches against customer standards — **good colour discrimination, not colour-blind**.
- **Batch adjustment to spec:** understood the practice of adding solvent stepwise to bring viscosity onto target, never below spec, since an over-thinned batch is hard to recover.
- **Production line familiarity:** high-speed disperser, basket mill and hydraulic let-down tank — understood how temperature, grinding time and resin feed rate affect batch quality.
- **Observed batch troubleshooting:** followed how plant engineers trace root causes of out-of-spec batches (resin shock from fast pumping, pigment flocculation, off-target density/viscosity/fineness) and adjust to recover a batch rather than scrap it.
- **Health, safety and environment:** followed fire-safety rules and PPE requirements, consulted MSDS, and segregated waste solvent and hazardous solid waste per plant procedure.

EDUCATION

B.Eng. in Chemical Engineering Technology

📅 2022 – 2026

Industrial University of Ho Chi Minh City (IUH) — Faculty of Chemical Engineering

📍 Ho Chi Minh City

- GPA **2.8/4.0**.
- **Graduation thesis:** *Factors affecting the production of composite and leather-like materials from water hyacinth, agricultural by-products and waste paper from garment embroidery*. Supervisor: Dr. Pham Thi Hong Phuong.
- Core coursework: Organic Chemistry, Physical Chemistry, Reaction Engineering, Unit Operations, Instrumental Analysis, Process Safety.

TECHNICAL SKILLS

Operated	Stormer viscometer, Pfund Cryptometer, Hegman gauge, density cup, gloss meter, adhesion tester
Exposure to	Salt spray chamber (ISO 7253), QUV weathering, Elcometer mandrel bend, impact tester; rust/blister rating ISO 4628-2, ASTM D1654
Lab instruments	Able to use standard measurement and basic analytical instruments in a university laboratory
Materials	Alkyd, acrylic, epoxy, polyurethane and polyester resins; solvent systems; pigments and extenders; natural-fibre composites (water hyacinth, agricultural by-products)
Quality systems	ISO 9001:2015, 5S, IQC/IPQC/FQC workflow, SOPs, navigating TCVN — ASTM — ISO
HSE	MSDS/SDS, PPE, fire safety, hazardous waste segregation and handling
Software	ChemDraw, Excel (test data handling), Word — technical reporting
Coating film testing	Applying standards
Chemical safety (HSE)	Lab instrumentation
Organic chemistry, materials	Data handling, records

RESEARCH & PROJECTS

● **Thesis: Leather-like composites from water hyacinth and agricultural by-products**  2026
Faculty of Chemical Engineering, IUH — Supervisor: Dr. Pham Thi Hong Phuong  Ho Chi Minh City

- Investigating the process factors governing leather-like material made from **water hyacinth, agricultural by-products and waste paper** from garment embroidery.
- Aimed at valorising waste streams into an environmentally friendlier substitute for synthetic leather.

● **Industrial internship report (team of 8)**  2026
Faculty of Chemical Engineering, IUH — Supervisor: Le Nhat Thong, M.Sc.  Ho Chi Minh City

- Documented the complete paint manufacturing process: raw materials, block flow diagram, equipment, common failures and corrective measures.
- Compiled **12 quality tests** with their principles, apparatus, governing standards and step-by-step procedures.

AWARDS

- **Second and Third Prize — Provincial Science & Engineering Fair** (social media category), high school.
- **Second Prize — Provincial Chess Championship**, high school.
- **National Consolation Prize — Safe Traffic for a Smiling Tomorrow** competition.
- **Provincial Consolation Prize — Reading Culture** competition.

CERTIFICATIONS

IT Certificate in Basic Information Technology Application — Industrial University of Ho Chi Minh City

OTHER ACTIVITIES

● **Volunteer Collaborator**  [dates]
Ward 5 People's Committee, [City/Province]  [location]

- Supported local community programmes and events; teamwork and direct engagement with residents.